Cyclotomic Gap Quotients, Recurrent GCD Signatures, and Finite-Field Synchronization

A nonduplicative continuation of the unified report on the Alaoglu–Erdős integer-exponent conjecture

22 August 2026

Problem. Prove that, for real x,

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z}.$$

Bottom line. I do not obtain a complete proof. I do obtain a new, exact cyclotomic dichotomy. Fix any nonzero lattice direction $(a, b) \in \mathbb{Z}^2$ and reduce $r = 2^a 3^b = A/B > 1$ and $r^x = M^a N^b = C/D > 1$. If x is a positive integer, then

$$\frac{C^n - D^n}{A^n - B^n} \in \mathbb{Z} \qquad (n \geq 1).$$

If x is a hypothetical noninteger solution, then instead

$$\log \operatorname{den}\left(\frac{C^n - D^n}{A^n - B^n}\right) = n \log A + o(n).$$

Thus essentially the entire exponential input gap survives in the reduced denominator, in every rational direction at once. This converts the conjecture into several precise finite-place targets: any fixed direction with a positive exponential denominator saving, any fixed finite prime support along infinitely many layers, any almost-everywhere support inclusion, or any bounded synchronization cost would finish the proof.
A second new package imports a June 2026 theorem on the sequence $\gcd(a^n - 1, b^n - 1)$. It yields a recurrence-signature characterization of integrality and shows exactly why constructing one nonperiodic common linear divisibility sequence would be a proof certificate. Finally, an exterior-power calculation shows that ordinary Hadamard sign preconditioning is globally determinant-neutral; any benefit must come from selective arithmetic structure, not orthogonal mixing itself.

Prepared from the attached unified report, but deliberately organized around objects and theorems absent from that report.

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1 Executive assessment

1.1 What is new in this continuation

The attached unified report already develops an exceptionally broad collection of methods: conditional free-monoid structure, primitive odd cores, finite verification, arbitrary-node and mixed-semigroup determinants, p-adic transport ceilings, Smith profiles, Loewner and Padé systems, rational secants between distinct lattice directions, Sturmian coding, canonical products, moment methods, and several exact sufficient conditions. Repeating any of those approaches would add volume rather than information.

This continuation therefore starts from three objects not developed there:

(i) the *same-direction power quotient*

$$Q_{\mathbf{a}}(n) = \frac{C_{\mathbf{a}}^n - D_{\mathbf{a}}^n}{A_{\mathbf{a}}^n - B_{\mathbf{a}}^n},$$

not a secant between two distinct exponent vectors;

(ii) the *source–output gcd sequence*

$$g_{2,M}(n) = \gcd(2^n - 1, M^n - 1),$$

and its constant-coefficient recurrence signature;

(iii) the *capture cost* $\kappa_{\mathbf{a}}(n)$, the least multiplier of a hypothetical solution needed to make its output reproduce the full finite-field torsion of an input gap at level n.

The resulting theorems use the Bugeaud–Corvaja–Zannier and Corvaja–Zannier gcd estimates in a way not present in the source: rather than recording them as a general obstruction to large gcds, we compute the exact exponential rate of the reduced denominator of every candidate gap quotient. The report also incorporates the June 2026 preprint of Nguyen-Dang on recurrent gcd sequences, which postdates much of the standard literature and is not treated in the attached manuscript.

1.2 Claim ledger

Statement	Status	Dependency
Directional gap-quotient dichotomy	**[proved here]**	Corvaja–Zannier's generalized S-unit gcd theorem, plus elementary height and prime-valuation arguments.
Recurrence signature of $g_{2,M}$	**[external theorem]**	Nguyen-Dang, arXiv:2606.07959 (2026), itself using the classical BCZ estimate and linear-divisibility-sequence theory.
Primitive-core collapse of the diagonal gcd sequences	**[proved here]**	Elementary congruences modulo $2^n - 1$ and $3^n - 1$.
Fixed-support, denominator-saving, and multiplier criteria	**[proved here]**	Immediate consequences of the directional dichotomy and LTE.

Statement	Status	Dependency
Support-inclusion criterion	[proved here]	Corrales-Rodrigáñez–Schoof's support problem theorem.
Exact synchronization-cost formula and divergence	[proved here]	Multiplicative orders, Euler's theorem, and the homogeneous gcd estimate.
Numerical tables	[exact computation]	Exact integer factorization and multiplicative orders; reproducible code is included.
Hadamard preconditioning neutrality	[proved here]	Orthogonal invariance of singular values and exterior powers.
Derivation of bounded synchronization from $2^x, 3^x \in \mathbb{Z}$	[open bridge]	This is the newly isolated archimedean-to-Frobenius bridge.
Complete proof of the conjecture	[open bridge]	No contradiction with a hypothetical counterexample has been obtained.

1.3 The strongest theorem in one line

For a positive solution x and every nonzero $\mathbf{a} = (a, b) \in \mathbb{Z}^2$, orient $r_{\mathbf{a}} = 2^a 3^b = A/B > 1$ and reduce $r_{\mathbf{a}}^x = M^a N^b = C/D > 1$. Then

$$x \in \mathbb{Z} \quad \Longrightarrow \quad \operatorname{den} Q_{\mathbf{a}}(n) = 1 \quad (n \geq 1),$$

whereas

$$x \notin \mathbb{Z} \quad \Longrightarrow \quad \frac{\log \operatorname{den} Q_{\mathbf{a}}(n)}{n} \longrightarrow \log A.$$

No intermediate asymptotic regime is possible.

This is not merely another equivalent formulation. It gives a quantitative rigidity law: a counterexample must be maximally decorrelated from the input at finite places, despite being perfectly correlated at the real place by the identity $M^a N^b = (2^a 3^b)^x$.

1.4 External-claims audit

The public mathematical record was checked because the prompt reports several very recent AI-assisted breakthroughs. The Jacobian claim has a publicly displayed explicit counterexample and substantial secondary discussion. The Hadamard claim is less settled: secondary posts assert that twelve formerly open orders below 2000 were filled, but current public Sage/open-search records still list order 668 as unresolved and I did not locate the claimed exact matrices. The rank-30 elliptic-curve claim is visible in social mirrors and aggregators, but I did not locate a primary paper or independent certificate with settled attribution. Most importantly for the present task, I found no public preprint, Lean repository, or checkable argument for a new Alaoglu–Erdős breakthrough as of 22 August 2026. Accordingly, no technical step from the rival team's private claim is assumed here.

Status: The news audit is evidentiary context, not a mathematical premise. The Hadamard and rank-30 claims should be treated as provisional until exact public certificates are available.

2 Minimal inherited setup

Let $x \in \mathbb{R}$ satisfy

$$M := 2^x \in \mathbb{Z}, \qquad N := 3^x \in \mathbb{Z}.$$

If $x < 0$, then $0 < 2^x < 1$, so no solution is possible. The case $x = 0$ is integral and trivial. Hence every counterexample would have $x > 0$ and $M, N \geq 2$.

We use only the following inherited facts from the unified report:

(i) a rational solution is an integer, by unique factorization;

(ii) every nonintegral solution is transcendental;

(iii) positive integer multiples of a solution are again solutions;

(iv) conditional on failure, one may choose a least nonintegral solution β with a primitive output normalization

$$M = 2^\beta = 2^a w^d,$$

where $w > 1$ is odd and power-primitive; symmetrically $N = 3^b v^e$, and $a = 0$ or $b = 0$;

(v) the full conditional solution set is the free monoid $\{j + k\beta : j, k \in \mathbb{N}_0\}$.

Only items (i) and (iii) are needed for most of this report. The primitive normalization is used to show that the new gcd and synchronization invariants ignore the base-adic depth and see only the primitive core.

Lemma 2.1 (Rational-exponent lemma). *If $q \in \mathbb{Q}$ and $2^q \in \mathbb{Z}$, then $q \in \mathbb{Z}$.*

Proof. Write $q = r/s$ in lowest terms with $s > 0$. If $2^{r/s} = m \in \mathbb{Z}$, then $m^s = 2^r$. Unique factorization gives $m = 2^t$ and $ts = r$. Since $\gcd(r, s) = 1$, we have $s = 1$. □

Lemma 2.2 (Output independence). *For every positive solution x, the integers M and N are multiplicatively independent.*

Proof. If $M^r = N^s$ for positive integers r, s, then $2^{rx} = 3^{sx}$. Since $x > 0$, this gives $2^r = 3^s$, impossible. □

3 Recurrent gcd signatures

3.1 The 2026 recurrence theorem

For integers $a, b \geq 2$, put

$$g_{a,b}(n) := \gcd(a^n - 1, b^n - 1) \qquad (n \geq 1).$$

A sequence is *C-finite* if it satisfies a constant-coefficient linear recurrence. It is a *linear divisibility sequence* (LDS) if it is both C-finite and satisfies $W_m \mid W_n$ whenever $m \mid n$.

Theorem 3.1 (Nguyen-Dang, 2026). *Let $a, b \geq 2$. The following are equivalent:*

(a) *a and b are multiplicatively dependent;*

(b) *$(g_{a,b}(n))_{n\geq 1}$ is C-finite;*

(c) *some tail of $(g_{a,b}(n))$ is C-finite.*

If a, b are multiplicatively independent, every integer LDS W_n satisfying

$$W_n \mid a^n - 1, \qquad W_n \mid b^n - 1 \qquad (n \geq 1)$$

is periodic.

Status: [external theorem] — Theorem 1.1 and Theorem 1.2 of Nguyen-Dang [6]. The preprint is recent and should be cited as a preprint, not as a peer-reviewed theorem.

3.2 A recurrence certificate for integrality

Theorem 3.2 (One-base recurrence signature)**.** *Let $x > 0$ and suppose $M = 2^x \in \mathbb{Z}$. The following are equivalent:*

(i) *$x \in \mathbb{Z}$;*

(ii) *2 and M are multiplicatively dependent;*

(iii) *$g_{2,M}(n)$ is C-finite;*

(iv) *some tail of $g_{2,M}(n)$ is C-finite;*

(v) *there exists a nonperiodic integer LDS W_n such that $W_n \mid 2^n - 1$ and $W_n \mid M^n - 1$ for all n.*

The analogous equivalence holds with $(3, N)$ in place of $(2, M)$.

Proof. If 2 and M are dependent, then $M^s = 2^r$ for some $r, s > 0$. Unique factorization forces $M = 2^k$ for an integer k, hence $x = k$. The converse is immediate. The equivalence with (iii)–(iv) is Theorem 3.1. If $x = k \in \mathbb{Z}_{>0}$, then $W_n = 2^n - 1$ is a nonperiodic LDS and divides $M^n - 1 = 2^{kn} - 1$. If $x \notin \mathbb{Z}$, Theorem 3.1 says that every such common LDS is periodic. □

Proof criterion 3.3 (LDS proof certificate)**.** *To prove the Alaoglu–Erdős conjecture it would suffice, from the simultaneous hypotheses $2^x, 3^x \in \mathbb{Z}$, to construct a single nonperiodic LDS dividing both $2^n - 1$ and $(2^x)^n - 1$ for every n.*

This criterion is much more rigid than merely constructing a divisibility sequence: the gcd sequence itself is always a divisibility sequence. The missing property is finite linear recurrence, or a nonperiodic recurrent factor.

3.3 The recurrence graph of the four integers

For $x > 0$ define a graph $\mathcal{R}_x$ on the vertices

$$\{2, 3, M, N\}$$

by joining a to b exactly when $g_{a,b}(n)$ is C-finite. By Theorem 3.1, this is exactly the multiplicative-dependence graph.

Theorem 3.4 (Recurrence-graph phase transition)**.** *Let $x > 0$ satisfy $M = 2^x, N = 3^x \in \mathbb{Z}$.*

(i) *If $x \in \mathbb{Z}$, then $\mathcal{R}_x$ has exactly the two edges $\{2, M\}$ and $\{3, N\}$.*

(ii) *If $x \notin \mathbb{Z}$, then neither of those two edges occurs. The only possible edges are $\{2, N\}$ and $\{3, M\}$, and at most one of them can occur.*

Proof. The source pair $(2, 3)$ and output pair (M, N) are independent, the latter by Lemma 2.2. The pairs $(2, M)$ and $(3, N)$ are dependent exactly when x is integral, by Theorem 3.2.

It remains to exclude simultaneous cross-dependence in the nonintegral case. If both cross edges occurred, unique factorization would give $N = 2^u$ and $M = 3^v$ for some positive integers u, v. With $\vartheta = \log 3/\log 2$, the equations $3^x = 2^u$ and $2^x = 3^v$ give $x\vartheta = u$ and $x = v\vartheta$, hence $\vartheta^2 = u/v \in \mathbb{Q}$. But ϑ is transcendental: rationality is impossible by unique factorization, while algebraic irrationality would contradict Gelfond–Schneider because $2^\vartheta = 3$. Thus both cross edges cannot occur. □

Remark 3.5. The graph theorem is a clean diagnostic, but not a proof: a generic hypothetical counterexample can simply have no recurrent edges. Its value is to state exactly what a successful recurrence construction must change.

3.4 Primitive-core collapse

The conditional normal form in the unified report becomes especially transparent modulo $2^n - 1$.

Lemma 3.6 (Dyadic depth is invisible)**.** *If $M = 2^a R$ with $a \geq 0$, then for every $n \geq 1$,*

$$\gcd(2^n - 1, M^n - 1) = \gcd(2^n - 1, R^n - 1).$$

In particular, if a least counterexample has $M = 2^a w^d$, then

$$g_{2,M}(n) = \gcd(2^n - 1, w^{dn} - 1).$$

Similarly, if $N = 3^b S$, then

$$g_{3,N}(n) = \gcd(3^n - 1, S^n - 1).$$

Proof. Modulo $2^n - 1$ one has $2^n \equiv 1$, so $M^n = 2^{an} R^n \equiv R^n$. The gcd identity follows. The base-3 statement is identical. □

Thus the new diagonal gcd sequence forgets the base-adic depth a or b and sees only the primitive base-free core. This explains, for example, why the exact sequences for $M = 3, 6, 12$ coincide.

4 Homogeneous gcds and directional gap quotients

4.1 The homogeneous S-unit gcd estimate

We need a rational version of the classical BCZ estimate.

Theorem 4.1 (Homogeneous gcd estimate)**.** *Let $A > B \geq 1$ and $C > D \geq 1$ be coprime pairs. If A/B and C/D are multiplicatively independent, then*

$$\log \gcd(A^n - B^n, C^n - D^n) = o(n).$$

Equivalently, for every $\varepsilon > 0$ the gcd is at most $\exp(\varepsilon n)$ for all sufficiently large n.

Derivation from Corvaja–Zannier. Apply Corollary 1 of Corvaja–Zannier [4] to the cyclic subgroup

$$\{((A/B)^n, (C/D)^n) : n \in \mathbb{Z}\} \subset \mathbb{G}_m^2(\mathbb{Q}).$$

Multiplicative independence keeps this subgroup outside every exceptional one-dimensional subtorus relevant to $(u-1, v-1)$. Their generalized gcd estimate gives sublinear logarithmic common finite-place mass. Because $(A, B) = (C, D) = 1$, a prime dividing A or B cannot divide $A^n - B^n$, and similarly for C, D. Hence every prime in the ordinary integer gcd is a nonexceptional finite place, and the generalized estimate is exactly the displayed $o(n)$ bound. □

Status: **[external theorem]** for the Subspace-Theorem input; **[proved here]** for the homogeneous specialization and all applications below. The original integer case $B = D = 1$ is the theorem of Bugeaud–Corvaja–Zannier [1].

4.2 A rational direction attached to a solution

Let $\mathbf{a} = (a, b) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$. Since 2 and 3 are multiplicatively independent, $2^a 3^b \neq 1$. Replace $\mathbf{a}$ by $-\mathbf{a}$ if needed and write in lowest terms

$$r_{\mathbf{a}} := 2^a 3^b = \frac{A_{\mathbf{a}}}{B_{\mathbf{a}}} > 1, \qquad (A_{\mathbf{a}}, B_{\mathbf{a}}) = 1.$$

For a positive solution x define

$$s_{\mathbf{a}} := M^a N^b = r_{\mathbf{a}}^x = \frac{C_{\mathbf{a}}}{D_{\mathbf{a}}} > 1, \qquad (C_{\mathbf{a}}, D_{\mathbf{a}}) = 1.$$

The identity $s_{\mathbf{a}} = r_{\mathbf{a}}^x$ follows simply from $M = 2^x, N = 3^x$ and remains valid for negative a or b.

Define the homogeneous gaps and their quotient by

$$U_{\mathbf{a}}(n) = A_{\mathbf{a}}^n - B_{\mathbf{a}}^n, \qquad V_{\mathbf{a}}(n) = C_{\mathbf{a}}^n - D_{\mathbf{a}}^n, \qquad Q_{\mathbf{a}}(n) = \frac{V_{\mathbf{a}}(n)}{U_{\mathbf{a}}(n)}.$$

Also put

$$G_{\mathbf{a}}(n) = \gcd(U_{\mathbf{a}}(n), V_{\mathbf{a}}(n)), \quad \Delta_{\mathbf{a}}(n) = \operatorname{den} Q_{\mathbf{a}}(n) = \frac{U_{\mathbf{a}}(n)}{G_{\mathbf{a}}(n)}.$$

Lemma 4.2 (Direction dependence detects rationality)**.** *For any nonzero direction $\mathbf{a}$, the rationals $r_{\mathbf{a}}$ and $s_{\mathbf{a}}$ are multiplicatively dependent if and only if $x \in \mathbb{Q}$. Under the hypothesis $M = 2^x \in \mathbb{Z}$, this is equivalent to $x \in \mathbb{Z}$.*

Proof. If $r^u = s^v$ with positive integers u, v, then $r^u = (r^x)^v$ and $r > 1$, so $x = u/v$. Conversely, if $x = u/v$, then $s^v = r^u$. The last assertion is Lemma 2.1. □

4.3 The directional denominator dichotomy

Theorem 4.3 (Directional gap-quotient dichotomy)**.** *Let $x > 0$ satisfy $M = 2^x, N = 3^x \in \mathbb{Z}$, and fix any nonzero direction $\mathbf{a} \in \mathbb{Z}^2$ oriented as above.*

(i) *If $x = k \in \mathbb{Z}_{>0}$, then*

$$Q_{\mathbf{a}}(n) = \frac{A_{\mathbf{a}}^{kn} - B_{\mathbf{a}}^{kn}}{A_{\mathbf{a}}^{n} - B_{\mathbf{a}}^{n}} \in \mathbb{Z} \qquad (n \geq 1).$$

In particular, $\Delta_{\mathbf{a}}(n) = 1$ for every n.

(ii) *If $x \notin \mathbb{Z}$, then*

$$\log G_{\mathbf{a}}(n) = o(n),$$

and

$$\begin{aligned} \log \Delta_{\mathbf{a}}(n) &= n \log A_{\mathbf{a}} + o(n), \\ \log \operatorname{num} Q_{\mathbf{a}}(n) &= n \log C_{\mathbf{a}} + o(n), \\ h(Q_{\mathbf{a}}(n)) &= n \max\{\log A_{\mathbf{a}}, \log C_{\mathbf{a}}\} + o(n). \end{aligned}$$

Proof. If $x = k \in \mathbb{Z}_{>0}$, reduction gives $C_{\mathbf{a}} = A_{\mathbf{a}}^k$ and $D_{\mathbf{a}} = B_{\mathbf{a}}^k$. The standard factorization $X^k - Y^k = (X - Y)(X^{k-1} + \cdots + Y^{k-1})$, with $X = A_{\mathbf{a}}^n$ and $Y = B_{\mathbf{a}}^n$, proves (i).

Suppose $x \notin \mathbb{Z}$. By Lemma 4.2, the ratios $A_{\mathbf{a}}/B_{\mathbf{a}}$ and $C_{\mathbf{a}}/D_{\mathbf{a}}$ are independent. Theorem 4.1 gives $\log G_{\mathbf{a}}(n) = o(n)$. Since

$$\log U_{\mathbf{a}}(n) = n \log A_{\mathbf{a}} + \log\left(1 - (B_{\mathbf{a}}/A_{\mathbf{a}})^n\right) = n \log A_{\mathbf{a}} + o(1),$$

we obtain

$$\log \Delta_{\mathbf{a}}(n) = \log U_{\mathbf{a}}(n) - \log G_{\mathbf{a}}(n) = n \log A_{\mathbf{a}} + o(n).$$

The numerator and height formulas are identical. □

Corollary 4.4 (One direction is enough)**.** *For a positive solution x and any fixed nonzero direction $\mathbf{a}$, the following are equivalent:*

(a) *$x \in \mathbb{Z}$;*

(b) *$Q_{\mathbf{a}}(n) \in \mathbb{Z}$ for every $n \geq 1$;*

(c) *$Q_{\mathbf{a}}(n) \in \mathbb{Z}$ for infinitely many unbounded n;*

(d) *$\log \Delta_{\mathbf{a}}(n) = o(n)$ along an unbounded sequence of n;*

(e) *for some $\delta > 0$, $\Delta_{\mathbf{a}}(n) \leq \exp((\log A_{\mathbf{a}} - \delta)n)$ for infinitely many n.*

Proof. The integral case follows from Theorem 4.3(i). Every one of (b)–(e) contradicts the limiting denominator rate in part (ii) unless x is integral. □

Remark 4.5 (Why this does not yet prove the conjecture). The original hypotheses give one exact integral value in some directions, but not infinitely many. For example,

$$\frac{M-1}{2-1} = M - 1 \in \mathbb{Z}.$$

The theorem says that a noninteger solution can accommodate any finite collection of such exceptions; the obstruction is asymptotic.

4.4 The four adjacent smooth directions

The prior report isolates the consecutive 3-smooth pairs $(1,2),(2,3),(3,4),(8,9)$. In the new language they correspond to the directions

$$2, \qquad \frac{3}{2}, \qquad \frac{4}{3}, \qquad \frac{9}{8}.$$

Their output ratios are respectively

$$M, \qquad \frac{N}{M}, \qquad \frac{M^2}{N}, \qquad \frac{N^2}{M^3}.$$

After reducing each output ratio to C/D, the denominator of $(C-D)/(A-B)$ is 1 because every input gap $A-B$ is 1.

Corollary 4.6 (One-step integrality followed by denominator explosion)**.** *If x is a hypothetical noninteger solution, then for each of the four directions above one has $Q(1) \in \mathbb{Z}$, but*

$$\log \operatorname{den} Q(n) = \begin{cases} n\log 2 + o(n), & r = 2, \\ n\log 3 + o(n), & r = 3/2, \\ n\log 4 + o(n), & r = 4/3, \\ n\log 9 + o(n), & r = 9/8. \end{cases}$$

This is a useful calibration: exact integrality at the first cyclotomic layer is automatic, while nearly maximal denominator growth is forced at high layers. Any proof based on the adjacent smooth pairs must therefore propagate $n = 1$ information to an unbounded set of power layers; local convexity or first-order gap information alone cannot do that.

5 Denominator-saving and cyclotomic criteria

The asymptotic formula in Theorem 4.3 becomes useful only after it is translated into conditions that a prospective argument might actually produce. This section records several such translations. They are elementary once the homogeneous gcd estimate is available, but they sharply delimit what remains to be proved.

5.1 A uniform valuation bound at fixed primes

Lemma 5.1 (Fixed-prime valuation bound)**.** *Let $A > B \geq 1$ be coprime integers and let p be a fixed prime. Then*

$$v_p(A^n - B^n) = O_{A,B,p}(\log(n+1)).$$

If $p \mid AB$, the valuation is identically zero.

Proof. If $p \mid A$, then $p \nmid B$ and $A^n - B^n \not\equiv 0 \pmod p$; the same argument applies when $p \mid B$. Assume now $p \nmid AB$. If no power $(A/B)^n$ is 1 modulo p, the valuation is again zero. Otherwise let $f = \operatorname{ord}_p(A/B)$. A nonzero valuation requires $f \mid n$. For odd p, LTE gives, after fixing the first admissible exponent f,

$$v_p(A^n - B^n) = v_p(A^f - B^f) + v_p(n/f).$$

For $p = 2$, the standard two-adic LTE formula has the same form up to an additive constant depending only on A and B. Since $v_p(n) \le \log n / \log p$, the assertion follows. □

The lemma says that a fixed finite set of primes can carry only polynomial, never exponential, mass from a binary power gap.

5.2 Equivalent one-direction proof targets

Fix a direction $\mathbf{a}$ and abbreviate

$$U_n = A^n - B^n, \qquad V_n = C^n - D^n, \qquad G_n = \gcd(U_n, V_n), \qquad \Delta_n = U_n/G_n.$$

Here $A/B = 2^a 3^b > 1$ and $C/D = (A/B)^x$ are in lowest terms.

Theorem 5.2 (A menu of finite-place proof criteria)**.** *Suppose $x > 0$ and $2^x, 3^x \in \mathbb{Z}$. If, for one fixed nonzero direction, any one of the following assertions holds along an unbounded sequence of indices n, then $x \in \mathbb{Z}$:*

(a) *there is a fixed finite set of primes S with* $\operatorname{supp}(\Delta_n) \subseteq S$*;*

(b) *there are positive integers K_n with $U_n \mid K_n V_n$ and $\log K_n = o(n)$;*

(c) *there is a $\delta > 0$ with $G_n \ge \exp(\delta n)$;*

(d) *there is a $\delta > 0$ with $\Delta_n \le \exp((\log A - \delta)n)$;*

(e) *the radical satisfies* $\log \operatorname{rad}(\Delta_n) = o(n/\log n)$ *and the exponents $v_p(\Delta_n)$ are uniformly $O(\log n)$ for primes in its support.*

Proof. Assume for contradiction that $x \notin \mathbb{Z}$. Then $\log \Delta_n = n \log A + o(n)$ and $\log G_n = o(n)$.

For (a), Lemma 5.1 gives

$$\log \Delta_n \le \sum_{p \in S} v_p(U_n) \log p = O_S(\log n),$$

a contradiction. For (b), the divisibility $U_n \mid K_n V_n$ implies $\Delta_n \mid K_n$: after cancelling G_n, the two integers U_n/G_n and V_n/G_n are coprime. Hence $\log \Delta_n \le \log K_n = o(n)$, again impossible. Conditions (c) and (d) directly contradict the two asymptotic formulas. Finally, (e) gives

$$\log \Delta_n = \sum_{p \mid \Delta_n} v_p(\Delta_n) \log p \le O(\log n) \sum_{p \mid \Delta_n} \log p = o(n),$$

again contradicting denominator saturation. □

Caution 5.3 (Radicals alone are delicate)**.** The bare estimate $\log \operatorname{rad}(\Delta_n) = o(n)$ is not by itself enough: a small collection of primes whose identities vary with n could carry large powers. A successful radical argument must control multiplicities as well. The fixed-prime lemma supplies such control only after the prime set has been stabilized.

Corollary 5.4 (Subexponential correction is fatal)**.** *For a hypothetical counterexample and every fixed direction,*

$$A^n - B^n \nmid K_n (C^n - D^n)$$

for all sufficiently large n whenever $K_n = \exp(o(n))$ is integral.

This is a useful target for approximation arguments. One need not force exact divisibility $U_n \mid V_n$; a multiplier of polynomial size, or even $\exp(o(n))$ size, would already contradict a counterexample.

5.3 Homogeneous cyclotomic layers

Write $\Phi_m(X, Y)$ for the homogeneous cyclotomic polynomial, characterized by

$$X^n - Y^n = \prod_{m \mid n} \Phi_m(X, Y).$$

Thus U_n is assembled from layers indexed by divisors of n. The denominator $\Delta_n = U_n/G_n$ measures exactly the prime-power mass of those layers that is not reproduced by the output gap V_n.

Proposition 5.5 (Collision-mass identity)**.** *For every n,*

$$\log G_n = \sum_p \min\{v_p(U_n), v_p(V_n)\} \log p,$$
$$\log \Delta_n = \sum_p \big(v_p(U_n) - \min\{v_p(U_n), v_p(V_n)\}\big) \log p.$$

If $x \notin \mathbb{Z}$, then the first sum is $o(n)$ and the second is $n \log A + o(n)$.

Proof. The identities are the prime factorizations of G_n and U_n/G_n; the asymptotics are Theorem 4.3. □

The proposition should be read as a conservation law. The input has total logarithmic mass $n \log A + o(n)$. In a counterexample, only $o(n)$ of that mass can collide with the output; almost all of it survives in the denominator.

Theorem 5.6 (Zsigmondy)**.** *Let $A > B \geq 1$ be coprime. Except for the classical cases*

(i) *$n = 2$ and $A + B$ is a power of 2;*

(ii) *$n = 6$ and $(A, B) = (2, 1)$,*

there is, for every $n > 1$, a prime dividing $A^n - B^n$ that divides no earlier $A^m - B^m$ with $1 \leq m < n$.

For the adjacent direction $A/B = 3/2$, neither exception occurs. Consequently every layer $n \geq 2$ introduces a genuinely new input prime. This observation alone is not enough: the output is allowed to acquire the same new prime at the same layer. The point of the gcd theorem is that such synchronized acquisitions can carry only $o(n)$ total logarithmic mass when the two ratios are independent.

Corollary 5.7 (Prime-index cyclotomic saturation)**.** *Let ℓ tend to infinity through primes. If $x \notin \mathbb{Z}$, then*

$$\log \operatorname{den}\left(\frac{C^\ell - D^\ell}{A^\ell - B^\ell}\right) = \ell \log A + o(\ell).$$

Since

$$A^\ell - B^\ell = (A - B)\Phi_\ell(A, B),$$

the fixed factor $A - B$ contributes only $O(1)$; asymptotically all of the denominator mass comes from the single new prime-index cyclotomic layer $\Phi_\ell(A, B)$.

Remark 5.8 (Why primitive divisors do not close the proof). Zsigmondy's theorem guarantees at least one fresh prime, not a large fresh factor. The fresh prime could be small compared with A^n. To finish by primitive divisors one needs a quantitative statement forcing a positive proportion of $\log \Phi_n(A, B)$ into primes or prime powers that the output cannot synchronize. Corollary 5.7 proves the aggregate failure of synchronization under the counterexample hypothesis, but it does not connect that failure back to the original integrality at level $n = 1$.

5.4 A useful contrapositive formulation

Proof criterion 5.9 (Cyclotomic propagation criterion)**.** *It would prove the Alaoglu–Erdős conjecture to derive, from $M = 2^x, N = 3^x \in \mathbb{Z}$, one direction A/B and an unbounded sequence n_j for which a positive fraction of the input gap is reproduced by the output:*

$$\log \gcd(A^{n_j} - B^{n_j}, C^{n_j} - D^{n_j}) \geq \delta n_j$$

for some fixed $\delta > 0$.

The desired fraction may be arbitrarily small. This is a substantially softer target than full divisibility, but it is still a genuinely finite-field statement: real approximation of $C/D = (A/B)^x$ does not by itself imply any common prime factors.

6 The support problem and unavoidable defect primes

The denominator dichotomy measures common prime-power mass. A logically independent route asks only which primes occur, ignoring valuations. A theorem of Corrales-Rodrigáñez and Schoof makes an almost-everywhere support inclusion completely rigid.

6.1 Support inclusion forces a power relation

For a reduced rational number $r = A/B > 1$ and a prime $p \nmid AB$, let $\mathrm{ord}_p(r)$ denote the multiplicative order of AB^{-1} in $\mathbb{F}_p^\times$. Then

$$p \mid A^n - B^n \quad \Longleftrightarrow \quad \mathrm{ord}_p(r) \mid n.$$

Thus inclusion between supports of all power gaps is equivalent to an order-divisibility condition at almost every prime.

Theorem 6.1 (Corrales–Rodrigáñez–Schoof)**.** *Let $r, s \in \mathbb{Q}^\times$ be nontorsion. Suppose that, for every positive integer n and all but finitely many primes p at which both numbers are units,*

$$r^n \equiv 1 \pmod p \quad \Longrightarrow \quad s^n \equiv 1 \pmod p.$$

Then $s = r^k$ for some integer k.

Status: [external theorem] — this is the multiplicative-group case of the support problem theorem [3].

Theorem 6.2 (Support-inclusion criterion for Alaoglu–Erdős)**.** *Fix one nonzero direction and write $r = A/B > 1$, $s = C/D = r^x > 1$. Either one of the following conditions implies $x \in \mathbb{Z}$:*

(a) *for every $n \geq 1$ and all but finitely many primes p,*

$$p \mid A^n - B^n \quad \Longrightarrow \quad p \mid C^n - D^n;$$

(b) *for every $n \geq 1$ and all but finitely many primes p,*

$$p \mid C^n - D^n \quad \Longrightarrow \quad p \mid A^n - B^n.$$

Proof. Condition (a) and Theorem 6.1 give $s = r^k$ for an integer k. Because $s = r^x$ and $r > 1$, one has $x = k$. Under (b), the theorem gives $r = s^k$, hence $x = 1/k \in \mathbb{Q}$. Lemma 2.1 then yields $x \in \mathbb{Z}$ (in fact $x = 1$ in this orientation). □

In particular, it is enough to prove the inclusion for the diagonal pair $(2, M)$ or $(3, N)$. No information from both bases is needed once almost-all-prime support containment has been obtained.

6.2 Defect sets forced by a counterexample

Define the directional defect sets

$$\mathcal{D}_{r\to s} = \{p : \text{ for some } n \geq 1,\ p \mid A^n - B^n,\ p \nmid C^n - D^n\},$$
$$\mathcal{D}_{s\to r} = \{p : \text{ for some } n \geq 1,\ p \mid C^n - D^n,\ p \nmid A^n - B^n\},$$

with the finitely many primes dividing $ABCD$ omitted.

Corollary 6.3 (Two-sided infinite support defects)**.** *If x is a hypothetical noninteger solution, then for every nonzero direction both $\mathcal{D}_{r\to s}$ and $\mathcal{D}_{s\to r}$ are infinite.*

Proof. If $\mathcal{D}_{r\to s}$ were finite, excluding it would give the support inclusion in Theorem 6.2(a), hence $x \in \mathbb{Z}$. The reverse direction is identical. □

This is stronger than saying the two gap sequences are not equal. Every rational direction must generate infinitely many distinct primes witnessing failure of support inclusion in each orientation.

Proposition 6.4 (Order-theoretic form of the defect)**.** *For $p \nmid ABCD$, the prime p belongs to $\mathcal{D}_{r\to s}$ exactly when*

$$\operatorname{ord}_p(s) \nmid \operatorname{ord}_p(r).$$

Similarly, $p \in \mathcal{D}_{s\to r}$ exactly when $\operatorname{ord}_p(r) \nmid \operatorname{ord}_p(s)$.

Proof. Take $n = \operatorname{ord}_p(r)$. Then $r^n \equiv 1 \pmod p$. The same exponent kills s precisely when $\operatorname{ord}_p(s) \mid \operatorname{ord}_p(r)$. This proves the first equivalence, and the second follows by symmetry. □

Consequently a counterexample must make the two order functions incomparable at infinitely many primes in both directions. The real identity $s = r^x$ imposes perfect proportionality on logarithms, but the reductions of r and s modulo primes must repeatedly destroy every integer divisibility relation between their orders.

6.3 A support certificate with a finite exceptional set

Proof criterion 6.5 (Finite-exception support certificate)**.** *A complete proof would follow from any argument producing a finite set of primes S such that, for one direction and every $n \geq 1$,*

$$\operatorname{supp}(A^n - B^n) \setminus S \subseteq \operatorname{supp}(C^n - D^n).$$

This target is qualitative and may be more accessible than the exponential gcd target of Criterion 5.9. On the other hand, it demands simultaneous control for every n. Zsigmondy supplies new input primes, but the original hypotheses do not say that the corresponding output order divides n. The next section quantifies the least multiplier needed to enforce that divisibility.

7 Finite-field capture cost

Support inclusion asks whether an input order kills the output. When it does not, there is still a least positive multiple of the input order that does. This least multiplier is the central new invariant of the report.

7.1 Removing the finitely many nonunit primes

Retain a direction $r = A/B > 1$, $s = C/D = r^x > 1$ in lowest terms. For $U_n = A^n - B^n$, define the CD-clean part

$$\widetilde{U}_n := \prod_{\substack{p^e \| U_n \\ p \nmid CD}} p^e.$$

Thus $\widetilde{U}_n$ is the largest divisor of U_n relatively prime to CD.

Lemma 7.1 (Cleaning costs only polynomial mass)**.** *One has*

$$\log \widetilde{U}_n = n \log A + O_{A,B,C,D}(\log(n+1)).$$

Proof. The full gap satisfies $\log U_n = n \log A + o(1)$. The quotient $U_n/\widetilde{U}_n$ is supported on the fixed finite set of primes dividing CD. Lemma 5.1 bounds the valuation at each such prime by $O(\log(n+1))$. □

The cleaning is indispensable. If a prime divides C or D, then CD^{-1} is not a unit modulo that prime power, so no multiplicative order is defined. Lemma 7.1 shows that deleting these places loses no exponential information.

7.2 Definition and exact order formula

Definition 7.2 (Full capture cost)**.** If $\widetilde{U}_n > 1$, let

$$t_n := \operatorname{ord}_{\widetilde{U}_n}(CD^{-1})$$

where D^{-1} is taken modulo $\widetilde{U}_n$. Define

$$\kappa_{r,s}(n) := \frac{t_n}{\gcd(t_n, n)}.$$

If $\widetilde{U}_n = 1$, set $\kappa_{r,s}(n) = 1$.

Proposition 7.3 (Exact least-multiplier interpretation)**.** *The integer $\kappa_{r,s}(n)$ is the least $k \geq 1$ such that*

$$\widetilde{U}_n \mid C^{kn} - D^{kn}.$$

Moreover,

$$1 \leq \kappa_{r,s}(n) \leq \lambda(\widetilde{U}_n) \leq \varphi(\widetilde{U}_n) \leq \widetilde{U}_n,$$

where λ is the Carmichael function.

Proof. The displayed divisibility is equivalent to $(CD^{-1})^{kn} \equiv 1 \pmod{\widetilde{U}_n}$, hence to $t_n \mid kn$. The least positive k with this property is $t_n/\gcd(t_n, n)$. The bounds follow from the definition of the multiplicative order and the standard inequalities $t_n \leq \lambda \leq \varphi \leq \widetilde{U}_n$. □

For a prime power $p^e \parallel \widetilde{U}_n$, put

$$\kappa_{p^e}(n) := \frac{\mathrm{ord}_{p^e}(CD^{-1})}{\gcd(\mathrm{ord}_{p^e}(CD^{-1}), n)}.$$

The Chinese remainder theorem gives the local factorization

$$\kappa_{r,s}(n) = \mathrm{lcm}_{p^e \parallel \widetilde{U}_n} \kappa_{p^e}(n).$$

At a primitive divisor p of $A^n - B^n$, one has $\mathrm{ord}_p(A/B) = n$; the local cost is exactly the excess in $\mathrm{ord}_p(C/D)$ not already absorbed by n.

Proposition 7.4 (Support and cost)**.** *For a fixed n, $\kappa_{r,s}(n) = 1$ exactly when every prime-power factor of $A^n - B^n$ outside the fixed set* $\mathrm{supp}(CD)$ *is reproduced by $C^n - D^n$. In particular, if $\kappa_{r,s}(n) = 1$ for infinitely many unbounded n, then $x \in \mathbb{Z}$.*

Proof. The first statement is Proposition 7.3 with $k = 1$. If it occurs infinitely often, then the reduced denominator of V_n/U_n is supported on the fixed set of primes dividing CD. Theorem 5.2(a) applies. □

7.3 The phase transition

Theorem 7.5 (Capture-cost dichotomy)**.** *Let $x > 0$ satisfy $2^x, 3^x \in \mathbb{Z}$, and fix a nonzero direction.*

(i) *If $x \in \mathbb{Z}$, then $\kappa_{r,s}(n) = 1$ for every $n \geq 1$.*

(ii) *If $x \notin \mathbb{Z}$, then*

$$\kappa_{r,s}(n) \longrightarrow \infty.$$

Equivalently, for every fixed K, all sufficiently large n require a multiplier larger than K to capture the cleaned input gap.

Proof. If $x = m \in \mathbb{Z}_{>0}$, then $C = A^m$, $D = B^m$ and $A^n - B^n \mid A^{mn} - B^{mn}$, so $k = 1$ is admissible.

Suppose $x \notin \mathbb{Z}$ and the costs fail to diverge. Then some K bounds $\kappa_{r,s}(n)$ on an infinite set. By the pigeonhole principle, a fixed $1 \le k \le K$ occurs for infinitely many n. For those n,

$$\widetilde{U}_n \mid C^{kn} - D^{kn},$$

so

$$\log \gcd(A^n - B^n, C^{kn} - D^{kn}) \ge \log \widetilde{U}_n = n \log A + O(\log n).$$

But A/B and $(C/D)^k$ are multiplicatively independent. Indeed, a dependence would give $r^u = s^{kv} = r^{xkv}$ for nonzero integers u, v, hence $x = u/(kv) \in \mathbb{Q}$, and then Lemma 2.1 would make x integral. The homogeneous gcd estimate applied to A/B and $(C/D)^k$ says that the logarithm of the same gcd is $o(n)$, a contradiction. □

The integrality transition is therefore exact:

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \kappa_{r,s}(n) \equiv 1 \quad \Longleftrightarrow \quad \liminf_{n\to\infty} \kappa_{r,s}(n) < \infty.$$

This holds in every fixed rational direction.

7.4 Partial capture cost

Full divisibility is stronger than needed. For $0 < \delta < \log A$, define, for all sufficiently large n,

$$\kappa_{r,s;\delta}(n) := \min \left\{ k \ge 1 : \log \gcd(A^n - B^n, C^{kn} - D^{kn}) \ge \delta n \right\}.$$

Existence follows because the full cost $k = \kappa_{r,s}(n)$ captures $\widetilde{U}_n$, whose logarithm is $n \log A + O(\log n)$.

Theorem 7.6 (Positive-mass capture also diverges)**.** *If $x \notin \mathbb{Z}$, then for every fixed $0 < \delta < \log A$,*

$$\kappa_{r,s;\delta}(n) \longrightarrow \infty.$$

If $x \in \mathbb{Z}$, then $\kappa_{r,s;\delta}(n) = 1$ for all sufficiently large n.

Proof. The integral case is immediate from $U_n \mid V_n$ and $\log U_n = n \log A + o(1)$. In the nonintegral case, boundedness on an infinite sequence would again produce one fixed k. The homogeneous gcd estimate for r and s^k would then give $o(n)$ common logarithmic mass, contradicting the defining lower bound δn. □

Thus even recovering one percent of the exponential input mass requires a multiplier tending to infinity. This is stronger than the qualitative support defect statement and weaker than full denominator saturation; it is often the most flexible target for a future argument.

7.5 Ray covariance and divisibility in the layer index

Proposition 7.7 (Covariance on a lattice ray)**.** *Let $m \ge 1$. If the direction $m\mathbf{a}$ is represented by the powered ratios $r^m = A^m/B^m$ and $s^m = C^m/D^m$, then*

$$Q_{m\mathbf{a}}(n) = Q_{\mathbf{a}}(mn), \qquad \kappa_{r^m,s^m}(n) = \kappa_{r,s}(mn).$$

Proof. Both statements reduce to the identities $(A^m)^n - (B^m)^n = A^{mn} - B^{mn}$ and $(C^m)^{kn} - (D^m)^{kn} = C^{kmn} - D^{kmn}$. The set of primes removed in the cleaning step is unchanged by taking powers. $\square$

Proposition 7.8 (A weak layer-divisibility inequality)**.** *If $m \mid n$ and $q = n/m$, then*

$$\kappa_{r,s}(m) \le q\, \kappa_{r,s}(n).$$

Proof. The cleaned divisor $\widetilde{U}_m$ divides $\widetilde{U}_n$. If $k = \kappa_{r,s}(n)$, then $\widetilde{U}_n$ divides $C^{kn} - D^{kn}$, so $\widetilde{U}_m$ divides $C^{kqm} - D^{kqm}$. Therefore qk is admissible at level m. $\square$

The inequality is too weak to force boundedness, but it is useful in computations and in a possible compactness argument on highly divisible indices.

7.6 Simultaneous synchronization of the two bases

Apply the construction to the diagonal directions

$$(r, s) = (2, M), \qquad (r, s) = (3, N),$$

and write $\kappa_2(n)$ and $\kappa_3(n)$. Set

$$\kappa_{\mathrm{syn}}(n) := \mathrm{lcm}(\kappa_2(n), \kappa_3(n)).$$

Theorem 7.9 (Two-base synchronization phase transition)**.** *For every positive solution x:*

(i) *if $x \in \mathbb{Z}$, then $\kappa_{\mathrm{syn}}(n) = 1$ for all n;*

(ii) *if $x \notin \mathbb{Z}$, then $\kappa_{\mathrm{syn}}(n) \to \infty$;*

(iii) *for every n,*

$$\kappa_{\mathrm{syn}}(n) < 6^n;$$

(iv) *with $L = \kappa_{\mathrm{syn}}(n)$, the same multiplier simultaneously gives*

$$\widetilde{(2^n - 1)}^{(M)} \mid M^{Ln} - 1,$$
$$\widetilde{(3^n - 1)}^{(N)} \mid N^{Ln} - 1,$$

where the tildes remove prime powers dividing M and N, respectively, and their logarithms are $n \log 2 + O(\log n)$ and $n \log 3 + O(\log n)$.

Proof. Parts (i) and (ii) follow from Theorem 7.5. The individual bounds $\kappa_2(n) \le 2^n - 1$ and $\kappa_3(n) \le 3^n - 1$ imply

$$\kappa_{\mathrm{syn}}(n) \le \kappa_2(n)\kappa_3(n) < 6^n.$$

Because L is a multiple of each individual least cost, both divisibilities in (iv) hold. The logarithmic sizes follow from Lemma 7.1. $\square$

Proof criterion 7.10 (Archimedean-to-Frobenius bridge)**.** *The Alaoglu–Erdős conjecture would follow if the simultaneous real relation*

$$\log M = x \log 2, \qquad \log N = x \log 3$$

could be used to prove that $\kappa_{\mathrm{syn}}(n)$ is bounded on an unbounded sequence of n. More generally, it is enough to bound both partial costs $\kappa_{2,M;\delta_2}(n)$ and $\kappa_{3,N;\delta_3}(n)$ for any fixed positive δ_2, δ_3.

This is the most concentrated open bridge found in the present work. The hypotheses are archimedean identities among logarithms; the conclusion concerns orders in finite multiplicative groups. Existing transcendence estimates control how closely integer powers can approach at the real place, but they do not directly bound these finite-field order multipliers.

7.7 Collapse to the primitive core

Proposition 7.11 (Base-adic depth is invisible to capture)**.** *If $M = 2^a R$ with $a \geq 0$, then the diagonal capture cost for $(2, M)$ equals that for $(2, R)$ at every layer. If $N = 3^b S$, then the cost for $(3, N)$ equals that for $(3, S)$.*

Proof. Modulo $2^n - 1$ one has

$$M^{kn} = 2^{akn} R^{kn} \equiv R^{kn}.$$

Also, because $2 \nmid 2^n - 1$, removing the prime factors of M removes exactly the prime factors of R that occur in $2^n - 1$. Hence the cleaned modulus and the admissible values of k are the same. The base-3 argument is identical. □

For a least hypothetical counterexample in the primitive normalization of the unified report, the synchronization invariant therefore sees only the odd/base-free cores. The large powers of 2 and 3 attached to those cores are arithmetically invisible at their own cyclotomic moduli.

8 Exact numerical reconnaissance

The theorems above are asymptotic and ineffective: the Subspace Theorem does not provide a practical threshold beyond which a chosen cost exceeds a chosen bound. Exact computation is therefore useful for calibrating the invariant, but it cannot certify the asymptotic regime. This section uses small independent pairs as surrogates and an integral pair as a control.

8.1 Control and surrogate pairs

For the input base 2 consider three outputs:

$$M = 4, \qquad M = 3, \qquad M = 5.$$

The first is the integral case $4 = 2^2$, so $\kappa_{2,4}(n) = 1$ identically. The other two are multiplicatively independent from 2 and therefore model the finite-field behavior that a noninteger solution would be forced to have in the diagonal direction. They are not claimed to extend to solutions of the simultaneous two-base problem.

Exact experiment 8.1 (Selected full-capture costs). Exact factorization of $2^n - 1$ and exact multiplicative orders give:

n	$\kappa_{2,4}(n)$	$\kappa_{2,3}(n)$	$\kappa_{2,5}(n)$
5	1	6	3
7	1	18	6
11	1	8	4
13	1	70	105
17	1	7710	3855
32	1	2048	2048
40	1	1542	1542
60	1	110	55

The independent costs are emphatically not monotone. For example, $\kappa_{2,3}(17) = 7710$ but $\kappa_{2,3}(60) = 110$. Theorem 7.5 says only that every fixed bound is eventually left behind; it does not assert monotonicity, and the ineffective threshold may be very large.

8.2 Local explanations for several rows

At prime indices with $2^n - 1$ prime, the cost is especially transparent:

$$\begin{aligned}
n = 5: &\quad 2^5 - 1 = 31, & \mathrm{ord}_{31}(3) = 30, &\quad \kappa_{2,3}(5) = 30/5 = 6; \\
n = 7: &\quad 2^7 - 1 = 127, & \mathrm{ord}_{127}(3) = 126, &\quad \kappa_{2,3}(7) = 126/7 = 18; \\
n = 13: &\quad 2^{13} - 1 = 8191, & \mathrm{ord}_{8191}(3) = 910, &\quad \kappa_{2,3}(13) = 910/13 = 70; \\
n = 17: &\quad 2^{17} - 1 = 131071, & \mathrm{ord}_{131071}(3) = 131070, &\quad \kappa_{2,3}(17) = 7710.
\end{aligned}$$

For $M = 4$, by contrast, $\mathrm{ord}_{2^n-1}(4)$ always divides n, because $4^n = 2^{2n} \equiv 1$; the cost is 1.

Composite layers show the lcm nature of the invariant. At $n = 32$,

$$2^{32} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537.$$

After deleting the factor 3 for the output $M = 3$, the order of 3 modulo the cleaned modulus is 65536, and hence

$$\kappa_{2,3}(32) = \frac{65536}{\gcd(65536, 32)} = 2048.$$

At $n = 60$, many local orders are already absorbed by the highly divisible layer index, which explains the much smaller cost 110.

8.3 Denominator ratios at finite height

For an output M define

$$\rho_M(n) := \frac{\log((2^n - 1)/\gcd(2^n - 1, M^n - 1))}{n \log 2}.$$

For multiplicatively independent 2 and M, the BCZ theorem gives $\rho_M(n) \to 1$, but convergence can be irregular. Some exact values, rounded only after the exact gcd computation, are:

n	$\gcd(2^n-1, 3^n-1)$	$\rho_3(n)$	$\gcd(2^n-1, 5^n-1)$	$\rho_5(n)$
11	23	0.588703	1	0.999936
17	1	0.999999	1	0.999999
32	85	0.799707	51	0.822737
40	11275	0.663479	1353	0.739951
60	47322275	0.575066	698476779	0.510339

The dips do not contradict the limit. Subexponential gcds can still be very large at moderate indices, and known lower-bound constructions show that large exceptional gcds occur infinitely often for independent bases. This warns against inferring asymptotics from a short scan.

8.4 What should be computed for an actual candidate

A hypothetical solution would supply the exact integers M and N. The most informative finite experiment would then be:

(i) compute $\kappa_2(n)$ and $\kappa_3(n)$ on prime, prime-power, and highly composite indices;

(ii) factor the local costs into excess order primes;

(iii) compute the partial costs for several fractions $\delta/\log 2$ and $\delta/\log 3$;

(iv) search for directions (a, b) of small height with anomalously small $\kappa_{\mathbf{a}}(n)$;

(v) test whether the same small multiplier recurs across both bases or across several adjacent smooth directions.

A bounded recurring multiplier would not merely be suggestive; by Theorem 7.5 it would be a rigorous contradiction certificate once an unbounded family of layers were proved.

8.5 Reproducible exact code

The following code generated Table 8.1. It uses no floating-point arithmetic in the factorization, order, gcd, or cost calculations.

Listing 1: Exact full-capture cost for a diagonal pair.

```
from __future__ import annotations

from math import gcd
from typing import Dict, Tuple

import sympy as sp


def cleaned_gap(base: int, output: int, n: int) -> Tuple[int, Dict[int, int]]:
    """Return the part of base**n - 1 coprime to output, with its factorization."""
    if base < 2 or output < 2 or n < 1:
        raise ValueError("base, output >= 2 and n >= 1 are required")

    gap = base**n - 1
```

```
    factors = sp.factorint(gap)
    modulus = 1
    kept: Dict[int, int] = {}
    for p, e in factors.items():
        if output % p != 0:
            modulus *= p**e
            kept[int(p)] = int(e)
    return modulus, kept


def full_capture_cost(base: int, output: int, n: int) -> int:
    """Least k >= 1 for which the cleaned gap divides output**(k*n) - 1."""
    modulus, _ = cleaned_gap(base, output, n)
    if modulus == 1:
        return 1

    order = int(sp.n_order(output % modulus, modulus))
    return order // gcd(order, n)


def verify_cost(base: int, output: int, n: int, k: int) -> bool:
    """Certify admissibility and minimality of a proposed full capture cost."""
    modulus, _ = cleaned_gap(base, output, n)
    if pow(output, k * n, modulus) != 1 % modulus:
        return False
    return all(pow(output, j * n, modulus) != 1 % modulus for j in range(1, k))


if __name__ == "__main__":
    indices = [5, 7, 11, 13, 17, 32, 40, 60]
    for n in indices:
        row = [n]
        for output in (4, 3, 5):
            k = full_capture_cost(2, output, n)
            assert verify_cost(2, output, n, k)
            row.append(k)
        print(*row)
```

For large n, complete factorization of $A^n - B^n$ becomes the bottleneck. The local formula permits partial certified lower bounds: factor any divisor $F \mid \widetilde{U}_n$, compute the order modulo each known prime power, and take the lcm of the resulting local costs. This lower bound is enough to rule out small synchronization multipliers without factoring the entire gap.

9 Hadamard preconditioning: what it can and cannot change

The prompt highlights recent Hadamard-matrix computations, so it is natural to ask whether a Hadamard sign transform could strengthen one of the determinant arguments in the unified report. There is a clean answer at the global level.

9.1 Analytic neutrality under orthogonal normalization

Let H be a Hadamard matrix of order m:

$$H \in \{\pm 1\}^{m\times m}, \qquad HH^{\mathsf{T}} = mI_m.$$

Then $U = m^{-1/2}H$ is orthogonal.

Theorem 9.1 (Exterior-power neutrality)**.** *Let A be a real $m \times n$ matrix and $1 \le k \le \min(m, n)$. Then:*

(i) *UA and A have the same singular values;*

(ii)

$$\| \wedge^k (UA)\|_{\mathrm{F}} = \| \wedge^k A\|_{\mathrm{F}};$$

(iii) *equivalently,*

$$\sum_{\substack{I\subseteq[m],\ |I|=k\\ J\subseteq[n],\ |J|=k}} \det((HA)_{I,J})^2 = m^k \sum_{\substack{I\subseteq[m],\ |I|=k\\ J\subseteq[n],\ |J|=k}} \det(A_{I,J})^2.$$

If A is square, then

$$|\det(HA)| = m^{m/2}|\det A|.$$

Proof. Left multiplication by the orthogonal matrix U preserves $A^{\mathsf{T}}A$, hence all singular values. The singular values of $\wedge^k A$ are the k-fold products of singular values of A, proving (ii). The squared Frobenius norm of $\wedge^k A$ is the sum of the squares of all $k \times k$ minors, giving (iii) after $H = \sqrt{m}\,U$. Finally, $|\det H| = m^{m/2}$ follows from $HH^{\mathsf{T}} = mI$. □

Thus Hadamard mixing can redistribute determinant mass among individual minors, but it cannot increase the aggregate exterior-power budget beyond the explicit factor $m^{k/2}$ already paid for by the row scaling. Any analytic determinant estimate based only on singular values, Frobenius norms, or the sum of squared minors is invariant after normalization.

9.2 p-adic neutrality away from the order

There is an equally useful arithmetic statement.

Theorem 9.2 (Local Smith-profile neutrality)**.** *Let A be an integer $m \times n$ matrix and H a Hadamard matrix of order m. For every prime $p \nmid m$, left multiplication by H preserves the Smith invariants of A over $\mathbb{Z}_p$. Equivalently, for every k, the p-adic valuation of the determinantal ideal generated by all $k \times k$ minors is the same for A and HA.*

Proof. Since

$$H^{-1} = \frac{1}{m} H^{\mathsf{T}},$$

the matrix H lies in $\mathrm{GL}_m(\mathbb{Z}_p)$ whenever $p \nmid m$. Multiplication by an invertible matrix over $\mathbb{Z}_p$ is a sequence of invertible row operations, which preserves the local Smith normal form and every Fitting/determinantal ideal. □

Corollary 9.3 (Only primes dividing the order can be globally gained)**.** *Any improvement in the common p-adic divisibility of all k-minors after Hadamard preconditioning must occur at a prime $p \mid m$. At primes $p \nmid m$, only the distribution among individual minors can change; the minimum valuation and the generated ideal cannot.*

This sharply limits the role of a newly available Hadamard order. Choosing m with many prime factors can create local room at those primes, but it also introduces the known global determinant factor $m^{m/2}$. A proof must exploit a specially selected minor or congruence, not generic orthogonal flattening.

9.3 A viable selective-minor program

The neutrality theorem does not make Hadamard matrices useless. It identifies the only remaining mechanism:

(i) form a structured interpolation or moment matrix A from the integer values $2^x, 3^x$ and their semigroup powers;

(ii) choose Hadamard rows so that one designated minor experiences systematic cancellation at the real place;

(iii) retain enough integrality that the same minor is divisible by a large known integer;

(iv) prove that this minor is nonzero and smaller in absolute value than its divisibility modulus.

The exterior-power identity says that any exceptionally small chosen minor is balanced by larger minors elsewhere. That is acceptable: a determinant contradiction needs one minor, not an average. The difficult step is to align the small analytic minor with a strong arithmetic divisor. No such alignment has yet been found.

Proof criterion 9.4 (Selective Hadamard certificate)**.** *A Hadamard-based proof must use information not invariant under left orthogonal action: a designated row subset, a sign-sensitive congruence, or a prime dividing the Hadamard order. Arguments depending only on the full determinant, singular values, or aggregate minor norms cannot improve the existing determinant balance.*

10 Attempts to close the synchronization bridge

The preceding sections isolate several exact proof certificates. This section records the main attempts made to derive one of them from the simultaneous integrality hypotheses, and pinpoints why each attempt presently stops. These are not excuses; they are constraints that a successful continuation must explicitly overcome.

10.1 Finite packets of directions

For $H \geq 1$, let

$$\mathcal{F}_H := \{(a,b) \in \mathbb{Z}^2 \setminus \{0\} : |a| + |b| \leq H\}/\{\pm 1\},$$

with each sign class oriented so that $2^a 3^b > 1$.

Proposition 10.1 (Uniform escape on every fixed packet)**.** *If x is a hypothetical noninteger solution, then for each fixed H,*

$$\min_{\mathbf{a} \in \mathcal{F}_H} \kappa_{\mathbf{a}}(n) \longrightarrow \infty.$$

The same assertion holds for every fixed positive-mass partial cost.

Proof. The packet is finite, and the cost in each fixed direction tends to infinity by Theorems 7.5 and 7.6. Given K, take the maximum of the finitely many thresholds beyond which each cost exceeds K. $\square$

Thus a counterexample cannot hide a permanently cheap small direction. A direction chosen adaptively with n could still have low cost, and the current Subspace-Theorem argument is not uniform when the height of that direction grows. One possible continuation is therefore a quantitative, height-uniform version of the directional gcd theorem.

Open question 10.2 (Growing-height uniformity)**.** *Can one prove a function $H(n) \to \infty$ for which a noninteger solution would force*

$$\min_{\mathbf{a}\in\mathcal{F}_{H(n)}} \kappa_{\mathbf{a}}(n) \to \infty?$$

A sufficiently rapid $H(n)$ could be confronted with geometry-of-numbers constructions of short lattice directions.

10.2 The three-gap triangle

The adjacent smooth nodes $1, 2, 3, 4$ produce three input gaps

$$U_0 = 2^n - 1, \qquad U_1 = 3^n - 2^n, \qquad U_2 = 4^n - 3^n.$$

They satisfy

$$U_0 + U_1 + U_2 = 4^n - 1, \qquad U_1 + U_2 = 2^n U_0.$$

Their output analogues are

$$V_0 = M^n - 1, \qquad V_1 = N^n - M^n, \qquad V_2 = M^{2n} - N^n,$$

with

$$V_0 + V_1 + V_2 = M^{2n} - 1, \qquad V_1 + V_2 = M^n V_0.$$

Proposition 10.3 (Exact pairwise overlap of the input triangle)**.** *For every $n \geq 1$,*

$$\gcd(U_0, U_1) = \gcd(U_0, U_2) = \gcd(U_1, U_2) = \gcd(2^n - 1, 3^n - 1).$$

Consequently the logarithm of every pairwise gcd is $o(n)$.

Proof. Modulo U_0 one has $2^n \equiv 4^n \equiv 1$, so both U_1 and $-U_2$ are congruent to $3^n - 1$. This proves the first two identities. Modulo U_1 one has $3^n \equiv 2^n$, hence

$$U_2 \equiv 4^n - 2^n = 2^n(2^n - 1).$$

The odd integer U_1 is coprime to 2^n, so $\gcd(U_1, U_2) = \gcd(U_1, 2^n - 1)$, which is the first gcd. The final estimate is BCZ for the independent bases 2 and 3. □

Corollary 10.4 (An almost-coprime gap packet)**.** *Let $L_n = \operatorname{lcm}(U_0, U_1, U_2)$. Then*

$$\log L_n = n \log 24 + o(n).$$

Proof. The three gaps have logarithms $n \log 2 + o(n)$, $n \log 3 + o(n)$, and $n \log 4 + o(n)$. Inclusion–exclusion for valuations, together with the $o(n)$ pairwise overlaps, gives the result; the triple overlap is bounded by a pairwise overlap. □

This is promising because the adjacent gaps jointly carry almost 24^n independent modulus. However, the telescoping output identities do not force any one input gap to divide its output counterpart. Writing $Q_i = V_i/U_i$ gives

$$Q_1U_1 + Q_2U_2 = M^nQ_0U_0,$$

but each product $Q_iU_i = V_i$ is already integral. The large, almost-coprime denominators are multiplied by precisely the gaps that cancel them, so the identity supplies no new cross-divisibility. A successful triangle argument needs an additional relation among the Q_i with coefficients that do not contain the full U_i.

10.3 Real near-collisions do not control finite-field orders

Take integers $u, v > 0$ and put

$$\varepsilon = u\log 2 - v\log 3.$$

Then

$$\begin{aligned} 2^u - 3^v &= 3^v(\mathrm{e}^{\varepsilon} - 1), \\ M^u - N^v &= N^v(\mathrm{e}^{x\varepsilon} - 1). \end{aligned}$$

Good rational approximations to $\log 3/\log 2$ make both relative gaps small, and the unified report extensively develops the resulting secant and determinant information. For the capture problem, however, one needs congruences such as

$$M^{kn} \equiv 1 \pmod{2^n - 1}.$$

The size of $M^u - N^v$ at the real place does not control its residue modulo a prime divisor of $2^n - 1$. Even an exponentially small relative error may correspond to an enormous nonzero integer difference containing none of the required primes.

There is one rigorous way a near-collision could help: prove that a product of desired input prime powers exceeds the absolute value of a nonzero integer known to be divisible by that product. Existing approximations do not achieve this balance. Their integer differences have exponential scale comparable to the moduli one wants to capture, and the known lower bounds for linear forms in logarithms prevent the extreme cancellation needed.

10.4 Why simultaneous bases have not yet bounded the cost

The two diagonal costs are built from

$$\mathrm{ord}_{p^e}(M) \quad (p^e \mid 2^n - 1), \qquad \mathrm{ord}_{q^f}(N) \quad (q^f \mid 3^n - 1).$$

The identity

$$\frac{\log M}{\log 2} = \frac{\log N}{\log 3} = x$$

links these integers only through the real embedding. There is no operation "raising to x" in the finite cyclic groups unless x is already known to be integral. In particular, $2^n \equiv 1 \pmod{p}$ does not imply any formal congruence for M^n.

A tempting Chinese-remainder construction chooses $L = \kappa_{\mathrm{syn}}(n)$, for which nearly all of both input gaps are captured at exponent Ln. This merely proves the existence of an $L < 6^n$; Euler's theorem already supplies such an exponential bound. The gcd estimate rules out bounded L, but not a growing L. Uniform estimates for

$$\gcd(2^n - 1, M^{Ln} - 1)$$

with both exponents varying would have to be strong enough to compare an $e^{\Theta(n)}$ gcd against a second exponent Ln. The qualitative Subspace Theorem estimate loses control as L grows.

10.5 Recurrence extraction: the exact obstruction

The gcd sequence $g_{2,M}(n)$ is automatically a divisibility sequence. One might hope that the simultaneous presence of $N = 3^x$ forces it to satisfy a recurrence. Theorem 3.1 shows that this hope, if realized, would immediately prove the conjecture; it also shows that there is no weaker nonperiodic LDS hidden inside the diagonal gcd under a counterexample.

The missing implication is therefore stark:

$$2^x, 3^x \in \mathbb{Z} \quad \overset{?}{\Longrightarrow} \quad g_{2,2^x}(n) \text{ has a nonperiodic recurrent factor.}$$

The local support of a gcd sequence is governed by multiplicative orders and is typically an infinite union of arithmetic progressions. A constant-coefficient recurrence has only finitely many characteristic roots. No mechanism is presently known that compresses the order data supplied by the second base into such a finite spectral set.

10.6 Primitive divisors: freshness without size

For every large layer, Zsigmondy provides a prime with new input order. At such a prime p, the local cost is

$$\frac{\mathrm{ord}_p(s)}{\gcd(\mathrm{ord}_p(s), n)}.$$

Corollary 6.3 ensures infinitely many instances where this exceeds 1; Theorem 7.5 forces unbounded aggregate excess. What is missing is a lower bound for the size or abundance of primitive primes with large excess output order. The standard primitive-divisor theorem is qualitative and allows one fresh prime to carry negligible logarithmic mass.

A potentially decisive strengthening would be:

> For one fixed direction, a positive proportion of $\log \Phi_n(A, B)$ is carried by prime powers p^e for which $\mathrm{ord}_{p^e}(C/D) \nmid Kn$ for every fixed K.

The current gcd theorem proves the aggregate conclusion for each fixed K, but only after summing over all layers; it does not tie it to a structure coming from the four original integer values.

10.7 Most promising next technical target

Among the available bridges, the following appears both sharp and narrowly formulated.

Conjecture 10.5 (Bounded synchronized positive-mass capture)**.** *If $M = 2^x$ and $N = 3^x$ are integers, then there exist $\delta_2, \delta_3 > 0$, a fixed K, and infinitely many n for which one can find $1 \le k \le K$ satisfying*

$$\log \gcd(2^n - 1, M^{kn} - 1) \ge \delta_2 n,$$
$$\log \gcd(3^n - 1, N^{kn} - 1) \ge \delta_3 n.$$

The conjecture is automatic with $k = 1$ when x is integral and impossible for a noninteger solution by Theorem 7.6. Proving it would settle Alaoglu–Erdős. Its advantage over the original statement is that it asks for a concrete finite-field phenomenon that can be attacked using multiplicative orders, Chebotarev-type distribution, or structured resultants.

11 Lean formalization blueprint

A formal development should not begin with the full transcendence statement. Most of the new work is an elementary finite arithmetic package that can be verified independently of the deep gcd theorems. This section gives a dependency-aware plan designed to expose the precise trusted boundary.

11.1 Proposed module structure

Module	Content
`GapDirection`	Reduced positive rational directions; homogeneous gaps $A^n - B^n$; powering a direction; covariance along a lattice ray.
`FixedPrimeValuation`	LTE-based bound $v_p(A^n - B^n) \le C + v_p(n)$, including the two-adic branch.
`CleanedGap`	Largest divisor of a gap coprime to a fixed integer; factorization and logarithmic-loss lemmas.
`CaptureCost`	Multiplicative order modulo the cleaned gap; least-multiplier formula; local lcm decomposition; elementary upper bounds.
`SupportDefect`	Equivalence between support inclusion and divisibility of finite-field orders; forward and reverse defect sets.
`GapTriangle`	Exact pairwise gcd identities for the nodes $1, 2, 3, 4$ and the lcm consequence, conditional on an abstract subexponential-gcd input.
`HadamardPrecondition`	Orthogonal normalization, determinant scaling, exterior-power norm identity, and local Smith invariance for primes not dividing the order.
`AEBridge`	Construction of A, B, C, D from (a, b) and a positive solution $M = 2^x, N = 3^x$; rationality detection from multiplicative dependence.
`ExternalGCD`	Clearly marked theorem interfaces for Corvaja–Zannier and Nguyen-Dang.
`Consequences`	Denominator saturation, capture divergence, fixed-support criteria, and recurrence signatures derived from the interfaces.

11.2 Core data structure

One convenient abstraction is:

Listing 2: Lean-style pseudocode for a reduced direction.

```
structure RatDirection where
  A B C D : Nat
  hB : 0 < B
  hD : 0 < D
  hAB : Nat.Coprime A B
  hCD : Nat.Coprime C D
  hABpos : B < A
  hCDpos : D < C

namespace RatDirection

def inputGap  (R : RatDirection) (n : Nat) : Nat := R.A^n - R.B^n
def outputGap (R : RatDirection) (n : Nat) : Nat := R.C^n - R.D^n

def cleanedGap (R : RatDirection) (n : Nat) : Nat :=
  (R.inputGap n).coprimePart (R.C * R.D)

-- Defined through the order of C * D^{-1} in ZMod (cleanedGap n).
def captureCost (R : RatDirection) (n : Nat) : Nat := ...

end RatDirection
```

The actual implementation may represent the unit CD^{-1} in $(\mathbb{Z}/\widetilde{U}_n\mathbb{Z})^\times$ rather than in the ambient ring, avoiding special cases in the order API.

11.3 Elementary theorem milestones

The first formal milestone should require no unproved number-theoretic interface:

L1. Least multiplier. Prove

$$\kappa(n) = \min\{k \geq 1 : \widetilde{U}_n \mid C^{kn} - D^{kn}\} = \frac{t_n}{\gcd(t_n, n)}.$$

L2. Local decomposition. Prove that the global cost is the lcm of its prime-power costs.

L3. Integral branch. If $C = A^m, D = B^m$, prove $\kappa(n) = 1$.

L4. Ray covariance. Prove $\kappa_{r^m,s^m}(n) = \kappa_{r,s}(mn)$.

L5. Layer inequality. For $m \mid n$, prove $\kappa(m) \leq (n/m)\kappa(n)$.

L6. Core collapse. Formalize the congruences $(2^a R)^{kn} \equiv R^{kn} \pmod{2^n - 1}$ and their base-3 analogues.

L7. Gap triangle. Prove all three pairwise gcd identities by modular rewriting.

L8. Hadamard local neutrality. Over $\mathbb{Z}_p$, construct the inverse $m^{-1}H^\mathsf{T}$ for $p \nmid m$ and transfer determinantal ideals.

These statements are exact and computationally testable. Completing them would already produce a reusable finite-field library independent of the status of Alaoglu–Erdős.

11.4 Deep theorem interfaces

The formal proof should isolate, not silently assume, three external results.

E1. Homogeneous gcd estimate. For multiplicatively independent reduced positive rationals $A/B, C/D$,

$$\log \gcd(A^n - B^n, C^n - D^n) = o(n).$$

A first Lean development can expose this as an axiomatically imported theorem with a precise bibliographic comment. A full formalization would require a major formal Subspace-Theorem project.

E2. Support problem. Almost-everywhere order divisibility implies an integral power relation. Formalizing the published proof would require algebraic number theory and Chebotarev machinery well beyond the elementary core.

E3. Recurrence classification. Nguyen-Dang's 2026 theorem should be a separate interface because it additionally depends on modern classification of integer linear divisibility sequences.

No final theorem should be labeled "fully formalized" while any of these interfaces remains an unverified axiom. The elementary consequences can nevertheless be machine-checked relative to clearly named assumptions.

11.5 Formal statement of the capture dichotomy

A practical theorem interface could look schematically as follows:

Listing 3: Lean-style pseudocode for the main conditional theorem.

```
axiom homogeneous_gcd_isLittleO
  (R : RatDirection)
  (hindep : MultiplicativelyIndependent (R.A / R.B) (R.C / R.D)) :
  IsLittleO atTop
    (fun n => Real.log (Nat.gcd (R.inputGap n) (R.outputGap n)))
    (fun n => (n : Real))

theorem captureCost_tendsto_atTop
  (R : RatDirection)
  (hindep : MultiplicativelyIndependent (R.A / R.B) (R.C / R.D)) :
  Tendsto R.captureCost atTop atTop := by
  -- contradiction: a bounded subsequence gives a fixed multiplier k;
  -- the cleaned input gap then lies in a forbidden exponential gcd.
  ...
```

The mathematical proof uses an infinite pigeonhole step. In Lean it may be cleaner to prove the contrapositive: for each fixed K, there exists N such that no $1 \le k \le K$ captures the cleaned gap for $n \ge N$, then take the maximum of the finitely many thresholds attached to $k = 1, \ldots, K$.

11.6 Certified computation

For concrete inputs, a certificate need not trust a factorization oracle. It can include:

(i) a prime-power factorization of a divisor $F \mid A^n - B^n$;

(ii) primality certificates for the listed primes;

(iii) modular exponentiation showing the claimed local order;

(iv) for each proper prime divisor of that order, a nonidentity power proving minimality.

The lcm of the certified local costs is then a formally verified lower bound for $\kappa(n)$. This is enough to establish that no multiplier up to K works, without a full factorization of the input gap.

11.7 What a final Lean theorem would still need

After all modules above, the ultimate statement would have the form

```
theorem alaoglu_erdos
  (x : Real)
  (h2 : IsInteger (Real.rpow 2 x))
  (h3 : IsInteger (Real.rpow 3 x)) :
  IsInteger x := by
  ...
```

The unresolved line remains the derivation of a bounded recurrence/support/capture certificate from `h2` and `h3`. Formalization will make this gap impossible to blur, which is a feature rather than a defect.

12 Conclusions and prioritized continuation plan

No complete proof has been obtained. The main progress is a change of coordinates that turns the conjecture into an exact finite-place phase transition.

> For every fixed rational lattice direction, integer exponents make every power-gap quotient integral and every capture cost equal to 1. A hypothetical noninteger solution forces the opposite extreme: the quotient denominator retains asymptotically all input exponential mass, support defects occur at infinitely many primes in both orientations, every fixed multiplier captures only $o(n)$ logarithmic mass, and the least full or positive-mass multiplier tends to infinity.

This package gives several independent ways to finish:

P1. Bounded positive-mass synchronization. Prove Conjecture 10.5. This is the highest-priority target because it asks only for a fixed positive fraction of two cyclotomic gaps, not full divisibility.

P2. Almost-everywhere support inclusion. Find a finite set S for which one diagonal support is contained in its output support outside S. The support theorem then finishes immediately.

P3. A nonperiodic common LDS. Use both integer outputs to construct any nonperiodic recurrent factor of $2^n - 1$ and $M^n - 1$. Nguyen-Dang's theorem turns this into a short certificate.

P4. Uniformity in growing directions. Quantify the homogeneous gcd estimate as the height of (a, b) grows, then combine it with a geometry-of-numbers choice of a short direction having unusually low capture cost.

P5. Selective Hadamard minor. Search only for sign patterns that align one arithmetically divisible minor with analytic cancellation. Do not spend effort on orthogonally invariant determinant norms, which are neutral by Theorem 9.1.

The decisive conceptual issue is no longer hidden inside a large interpolation determinant. It is the absence of a theorem transmitting a common real logarithmic scale to bounded multiplicative-order distortion in finite fields. Any claimed breakthrough should be checked against this bridge: it must produce recurrence, support inclusion, exponential gcd mass, or bounded capture in at least one fixed direction.

Status: Final status on 22 August 2026: substantial new conditional rigidity and several exact proof certificates; no proof of the Alaoglu–Erdős conjecture.

A Nonduplication map relative to the unified report

The attached source is a long unified manuscript incorporating twenty-seven continuations. The present report was deliberately built around terminology and constructions absent from that source. A literal search of the supplied `.tex` found no occurrence of "Zsigmondy", "primitive divisor", "support problem", "linear recurrence", "Ailon", "synchronization cost", "Hadamard matrix", or "preconditioning".

More importantly, the mathematical objects differ as follows.

New object in this report	Nearest object in the unified report	Essential difference
Same-direction quotient $Q_{\mathbf{a}}(n)$	Rational secants between distinct lattice nodes	Here one fixes a single multiplicative direction and raises both rational ratios to the same layer n; cyclotomic divisibility, not interpolation geometry, is central.
Denominator saturation	Denominators of transferred convergents/secants	The new rate is an exact asymptotic $n \log A + o(n)$ obtained from a homogeneous generalized gcd theorem in every direction.
Recurrence signature of $g_{2,M}$	General references to BCZ-type gcd bounds	The 2026 classification detects integrality by C-finiteness and rules out every nonperiodic common LDS under independence.
Support defects	Prime valuations in determinant entries	The support theorem ignores multiplicity and converts almost-all-prime order inclusion directly into a power relation.

New object in this report	Nearest object in the unified report	Essential difference
Capture cost $\kappa(n)$	p-adic transport ceilings	Capture cost is a multiplicative order quotient: the exact least exponent multiplier needed to reproduce a whole cleaned cyclotomic gap.
Three-gap triangle	Adjacent smooth-node secants	The new statement is an exact pairwise gcd identity and an exponential lcm computation for the three input gaps.
Hadamard sign preconditioning	Entrywise/Hadamard powers of matrices	A Hadamard matrix acts by orthogonal row mixing; the report proves analytic exterior-power neutrality and p-adic Smith neutrality away from primes dividing its order.
Formalization plan	Informal proof-roadmap sections	The Lean plan splits elementary, computationally certifiable arithmetic from three explicit deep theorem interfaces.

Only the minimal inherited facts listed in Section 2 were reused. They are necessary to connect the new invariants to the original conjecture and were not reproved at the scale of the source report.

B Dependency graph and trust boundary

The logical flow is:

$$\begin{array}{c}\text{Corvaja–Zannier generalized gcd theorem}\\ \Downarrow\\ \text{homogeneous gcd estimate}\\ \Downarrow\\ \text{directional denominator saturation}\\ \Downarrow\\ \text{fixed-support criteria} \qquad \text{capture divergence} \qquad \text{positive-mass divergence}\end{array}$$

Separately,

$$\text{Corrales–Schoof support theorem} \Longrightarrow \text{support-inclusion certificate,}$$

and

$$\text{Nguyen-Dang recurrence theorem} \Longrightarrow \text{recurrence/LDS certificate.}$$

The gap triangle, exact capture formula, ray covariance, core collapse, numerical certificates, and Hadamard neutrality are elementary and do not depend on any unproved conjecture. The only unresolved implication is from the original simultaneous integrality hypotheses to one of the displayed certificates.

C Proof details omitted from the main flow

C.1 Why multiplicative independence survives a fixed multiplier

If $s = r^x$ with $r > 1$ and x is a hypothetical noninteger solution, then r and s^k are multiplicatively independent for every fixed nonzero integer k. Indeed, a relation $r^u = (s^k)^v$ gives $u = xkv$, hence $x = u/(kv) \in \mathbb{Q}$; the rational-exponent lemma then makes x integral. This observation is used every time the homogeneous gcd theorem is applied to $C^{kn} - D^{kn}$.

C.2 The local lcm formula

Let t_i be the order of s modulo the prime-power factors of $\widetilde{U}_n$. The order modulo the full cleaned modulus is $t = \mathrm{lcm}_i\, t_i$. For every prime q,

$$v_q\left(\frac{t}{\gcd(t,n)}\right) = \max\{0, \max_i v_q(t_i) - v_q(n)\} = \max_i v_q\left(\frac{t_i}{\gcd(t_i,n)}\right).$$

This proves

$$\frac{t}{\gcd(t,n)} = \mathrm{lcm}_i \frac{t_i}{\gcd(t_i,n)}.$$

C.3 A corrected radical criterion

A quantitatively safe version of Theorem 5.2(e) is the following. If along an unbounded sequence

$$\log \mathrm{rad}(\Delta_n) = o\left(\frac{n}{\log n}\right)$$

and every $v_p(\Delta_n) = O(\log n)$ with an absolute implied constant, then $\log \Delta_n = o(n)$ and x is integral. The factor $1/\log n$ cannot be omitted without a stronger multiplicity estimate.

D Research provenance and literature notes

The attached unified report was read as the baseline, not merely used as a source of the problem statement. The recent-literature search then focused on gcd sequences, support problems, primitive divisors, and verifiable public records of the breakthroughs mentioned in the prompt.

Nguyen-Dang's June 2026 preprint is the only post-2025 theorem used substantively. Its main recurrence and common-LDS statements are imported exactly as external results; its recent status is explicitly marked. The denominator theorem relies on the established Corvaja–Zannier generalized gcd estimate. The support criterion relies on the 1997 theorem of Corrales-Rodrigáñez and Schoof.

For the news audit, an August 2026 arXiv paper gives a self-contained account of the announced three-dimensional Jacobian counterexample. By contrast, public Hadamard construction records available during this work still listed order 668 as unresolved. This discrepancy is why the report uses Hadamard matrices only conditionally and does not rely on the prompt's claim that all twelve missing orders below 2000 have been constructed.

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